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RESEARCH INTERESTS

My research interests include: spatio-temporal data management and analysis, blockchain technology, and energy management.

EDUCATION

Northeastern University, China <i>PhD</i> Major: Computer Science and Technology	Sep. 2020 – Jan. 2026 Shenyang, China
Aalborg University, Denmark <i>Guest PhD</i> Major: Computer Science	Dec. 2023 – Dec. 2025 Aalborg, Denmark
Northeastern University, China <i>Master</i> Major: Biomedical Engineering	Sep. 2018 – Jul. 2020 Shenyang, China
Northeastern University, China <i>Bachelor</i> Major: Computer Science and Technology	Sep. 2013 – Jul. 2017 Shenyang, China

WORKING EXPERIENCE

Postdoc <i>Aalborg University, Denmark</i>	Jan. 2026 – Jul. 2026
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PUBLICATIONS

[†] denotes the corresponding author

Accepted:

1. (CORE A*) Qiuyang Li, Qian Ma, **Zhongming Yao**, Shikai Guo, Yu Gu, Ge Yu, Tianyi Li. *ReCATS: Replay-Free Continual Anomaly Detection for Non-Stationary Multivariate Time Series*. ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2026.
2. (CORE A*) Qian Ma, Linfei Dai, **Zhongming Yao**, Yu Gu, Tianyi Li, Christian S. Jensen, Ge Yu. *Multimodal Knowledge Graph Completion via Relation-Aware Negative Sampling with Diffusion-based Interpolation*. The International Conference on Very Large Databases (PVLDB), 2026.
3. (CORE A*) Fengyu Li, Junhao Zhu, Kaishi Song, Lu Chen, **Zhongming Yao**, Tianyi Li, Christian S. Jensen. *Replacing Multi-Step Assembly of Data Preparation Pipelines with One-Step LLM Pipeline Generation for Table QA*. The International Conference on Very Large Databases (PVLDB), 2026.
4. (CORE A*) Wentao Xu, **Zhongming Yao**[†], Weihao Li, Zhenghang Song, Tianyi Li, Yushuai Li. *TCRL: Temporal-Coupled Adversarial Training for Robust Constrained Reinforcement Learning in Worst-Case Scenarios*. International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2026.
5. (CORE A*) Mo Li, Zhaosong Zhao, Linlin Ding, Renata Borovica-Gajic, **Zhongming Yao**, Jianxin Li. *One-for-All Community Search on Unseen Graphs*. International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2026.

6. Jiachen Xu, Torben Bach Pedersen, **Zhongming Yao**, Tianyi Li, Yushuai Li. *FOgym: A Bottom-Up Home Energy Management Framework Based on FlexOffers and Multi-Agent Reinforcement Learning*. ACM International Conference on Future and Sustainable Energy Systems (**e-Energy**), 2026.
7. (**CORE C**) Yusen Mao, **Zhongming Yao**, Junchang Xin, Chenchen Yan, Yumeng Song, Zhongzhong Jiang, Zhiqiong Wang. *EVT²: Efficient and Verifiable Trajectory Top-k Queries in Hybrid-Storage Blockchains*. APWeb-WAIM joint International Conference on Web and Big Data (**APWeb-WAIM**), 2026.
8. Chenchen Yan, **Zhongming Yao**, Zhiqiong Wang, Kun Hao, Yusen Mao, Yushuai Li, Tianyi Li, Junchang Xin. *FlexBC: A Flexible Blockchain Storage Mechanism Based on Data Hotness Features through Off-Chain Offloading*. International Conference on Blockchain, Artificial Intelligence, and Trustworthy Systems (**BlockDys**), 2026.
9. (**CORE A***, **JCR Q1**) Qi Shen, Junchang Xin, Bingtian Dai, **Zhongming Yao**, Xinyao Liu, Zhiqiong Wang. *Towards Heterogeneous Multimodal Sentiment Analysis with ASR Errors via Multi-Granularity Contrastive Coordination*. IEEE Transactions on Multimedia (**TMM**), 2026.
10. (**JCR Q1**) Jiaqi Tian, Bonan Huang, Tianyi Li, David Wenzhong Gao, **Zhongming Yao**, Yumeng Song, Yushuai Li. *Digital Twin for Multitimescale Economic Dispatch in Cyber-Physical Energy Systems*. IEEE Transactions on Industrial Informatics (**TII**), 2025.
11. (**JCR Q1**) Xiaoyu Zhang, Qiuye Sun, Tianyi Li, Yumeng Song, **Zhongming Yao**, Yushuai Li. *A Survey on Large Language Models Enhanced Reinforcement Learning for Smart Grid*. Journal of Modern Power Systems and Clean Energy (**MPCE**), 2026.
12. Wentao Xu, Yushuai Li, **Zhongming Yao**, Tianyi Li, Zhenghang Song and Yang Zhang. *MiRoCPO: Mutual Information and RoCoF-Aware Constrained Policy Optimization for Robust Load Frequency Control Against Temporal-Coupled Attacks*. Cyber-Physical Energy Systems (**CPES**), 2026.
13. (**CORE A***) **Zhongming Yao**, Tianyi Li, Junchang Xin, Yushuai Li, Chenxu Wang, Zhiqiong Wang, Divesh Srivastava, Christian S Jensen. *VGQ: Enabling Verifiable Graph Queries on Blockchain Systems*. The IEEE International Conference on Data Engineering (**ICDE**), 2025.
14. (**JCR Q1**) Ziming Liu, Cheng Zhang, **Zhongming Yao**, Tianyi Li, Qiuye Sun, Yushuai Li. *Clustered Federated Learning for Energy-Harvesting Smart Meters in P2P Energy Trading*. IEEE Transactions on Green Communications and Networking (**TGCN**), 2025.
15. (**JCR Q1**) Tianyi Li, Yushuai Li, Yumeng Song, **Zhongming Yao**, Wei Gao, David Wenzhong Gao. *Networked Digital Twins for Autonomous Vehicles: A New Perspective*. IEEE Transactions on Intelligent Vehicles (**TIV**), 2024.
16. (**CORE C**) **Zhongming Yao**, Zhiqiong Wang, Kun Hao, Liang Wen. *Efficient Blockchain Data Provenance Based on the W3C PROV Model*. The International Conference on Advanced Data Mining and Applications (**ADMA**), 2023.
17. (**CORE C**) Liang Wen, Zhiqiong Wang, Tingyu Cui, Caiyun Shi, Baoting Li, **Zhongming Yao**. *A Fine-Grained Verification Method for Blockchain Data Based on Merkle Path Sharding*. The International Conference on Advanced Data Mining and Applications (**ADMA**), 2023.
18. (**JCR Q1**) **Zhongming Yao**, Junchang Xin, Kun Hao, Zhiqiong Wang, Wancheng Zhu. *Learned Index Based Semantic Keyword Query on Blockchain*, Mathematics, 2023.
19. (**JCR Q1**) Kun Hao, Junchang Xin, Zhiqiong Wang, **Zhongming Yao**, Guoren Wang. *Efficient and Secure Data Sharing Scheme on Interoperable Blockchain Database*. IEEE Transactions on Big Data (**TBD**), 2023.
20. (**CORE B**, **JCR Q3**) Kun Hao, Junchang Xin, Zhiqiong Wang, **Zhongming Yao**, Guoren Wang. *On efficient top-k transaction path query processing in blockchain database*. Data & Knowledge Engineering (**DKE**), 2022.

21. Sihan Dong, Junchang Xin, Kun Hao, **Zhongming Yao**, Jinyi Chen. *A join query optimization algorithm in multi-blockchain environment*. Journal of Zhejiang University (Engineering Science), 2021. (In Chinese)

Ongoing papers:

1. (CORE A*) **Zhongming Yao**, Yan Zhao, Yumeng Song, Jiannan Guo, Lei Jia, Guangqiang Yin, Tianyi Li, Christian S. Jensen. *Verifiable and Privacy-Preserving Spatial Crowdsourcing*. The International Conference on Very Large Databases (PVLDB), 2027. (Under review)
2. (CORE A*) **Zhongming Yao**, Junchang Xin, Yumeng Song, Yusen Mao, Kristian Torp, Divesh Srivastava, Yushuai Li, Christian S. Jensen, Tianyi Li. *VTRQ: Enabling Verifiable Trajectory Range Queries in Hybrid-Storage Blockchains*. The VLDB Journal (VLDBJ), 2026. (Under review)
3. (CORE A*) Zongye Tao, Qian Ma, Shikai Guo, **Zhongming Yao**, Yu Gu, Ge Yu, Divesh Srivastava, Tianyi Li. *MIRA-D: Missingness-Informed Robust Alignment and Dependency Recovery for Time-Series Imputation*. The International Conference on Very Large Databases (PVLDB), 2027. (Under review)
4. (CORE A*) Yumeng Song, Tianyi Li, **Zhongming Yao**, Yushuai Li, Kristian Torp, Christian S. Jensen, Ira Assent. *Graph Contrastive Clustering for Traffic Sign Geolocalization*. IEEE International Conference on Data Mining (ICDM), 2026. (Under review)
5. (JCR Q1) Mingcan Wang, Junchang Xin, **Zhongming Yao**, Kaifu Long, Xian Zhang, Zhiqiong Wang. *GNN-MPPR: Enhancing the Propagation of Graph Neural Networks via Motif-Based Personalized PageRank*. IEEE Transactions on Network Science and Engineering (TNSE), 2026. (Under review)
6. Xinyang Li, Zhihong Cui, **Zhongming Yao**, Zhangkai Wu, Hengyu Liu, Kristian Torp, Minghe Yu, Tiancheng Zhang, Ge Yu. *StyleTimeQAM: Modeling Learning Style and Temporal Dynamics for Question-Answer Matching in Online Teaching Groups*. International Conference on Behavioural and Social Computing (BESC), 2026. (Under review)
7. (CORE A) Keqi Zhou, Rui Zhou, Junchang Xin, **Zhongming Yao**, Zhiqiong Wang, Zhongzhong Jiang, Tianyi Li. *Intimate Community Search in Weighted Bipartite Graphs*. International ACM Conference on Knowledge and Information Management (CIKM), 2026. (Under review)
8. (CORE B, JCR Q1) Feng Zhou, Junchang Xin, Bing Tian Dai, Qi Shen, Shudi Zhang, **Zhongming Yao**, Zhiqiong Wang. *Adaptive Conditional Perception Guided RWKV Network for Multi-Degradation LDCT Image Denoising*. Expert Systems With Applications (ESWA), 2026. (Under review)
9. (JCR Q2) Feng Zhou, Junchang Xin, Bing Tian Dai, Qi Shen, Shudi Zhang, **Zhongming Yao**, Zhiqiong Wang. *SSGR-Net: Symmetric semantic guided receptance weighted key-value framework for LDCT image denoising*. Biomedical Signal Processing and Control (BSPC), 2026. (Under review)
10. (CORE A*, JCR Q1) Chenchen Yan, **Zhongming Yao**, Junchang Xin, Yumeng Song, Tianyi Li, Yushuai Li, Zhiqiong Wang. *Verifiable kNN Queries on Multimodal Medical Data in Hybrid-Storage Blockchain Systems*. IEEE Journal of Biomedical and Health Informatics (JBHI), 2026. (To be submitted)
11. (CORE A*) Yumeng Song, Bingjie Li, Tianyi Li, Yu Gu, **Zhongming Yao**, Yushuai Li, Christian S. Jensen, Ge Yu. *TrajLite: A Query-Aware Trajectory Data Management System*. The IEEE International Conference on Data Engineering (ICDE), 2027. (To be submitted)

RESEARCH EXPERIENCE

- * **Research on Verifiable Spatial Crowdsourcing.** The research aims to support verifiable and privacy-preserving task assignment in spatial crowdsourcing by designing an atomic-fact commitment framework over RSA accumulators, which yields compact fairness and completion proofs with constant on-chain storage and no leakage of other workers' information.

- * **Research on Trajectory Range Queries.** The research aims to support efficient and trustworthy verifiable range queries over outsourced trajectory data by designing a novel and optimized authenticated data structure for managing trajectory data in hybrid blockchain systems.
- * **Research on Verifiable Graph Queries.** The research aims to improve the execution efficiency of graph queries by integrating blockchain systems with external graph databases, and to enable verifiable graph queries by designing a result verification method, thereby balancing performance and trustworthiness.
- * **Research on Provenance Data Management.** This research aims to develop a trustworthy framework for provenance data management, leveraging a blockchain system to record and maintain data version histories, thereby ensuring verifiability while enhancing query performance.

PROJECT EXPERIENCE

- * **Fair Task Assignment with Worker Preferences and Supply-demand Prediction in Spatial Crowdsourcing (Inge Lehmann Programme).** The project focuses on online spatial crowdsourcing for effective task assignment in transportation platforms, making three main contributions: (i) modeling workers' temporal preferences; (ii) designing adaptive and fairness-aware assignment methods; and (iii) predicting future supply and demand and integrating them into a reinforcement-learning-based assignment framework.
- * **Research on Inter-Cloud Element Data Management and Analysis.** The project focuses on achieving cloud supervision and governance through the management and analysis of inter-cloud element data, making three main contributions: (i) proposing mechanisms for multi-party negotiation and dynamic inter-cloud resource reconfiguration; (ii) developing methods for inter-cloud workflow scheduling; and (iii) designing a federated governance framework to enable large-scale cross-domain collaboration.
- * **Research on Provenance Data Management and Analysis in Large-Scale Manufacturing Systems.** The project focuses on achieving trustworthy traceability in large-scale manufacturing systems, and proposes three approaches: (i) constructing a blockchain-based collaborative sharing mechanism to improve data sharing; (ii) enhancing a sharding-based consensus mechanism to support cross-domain data exchange; and (iii) developing privacy-preserving methods for data supervision throughout the product life cycle.
- * **Research on Large-Scale Medical Data Management.** The project focuses on the management of large-scale medical data and proposes three main technical contributions: (i) constructing a hierarchical blockchain model to improve data access and sharing efficiency across heterogeneous sources; (ii) designing a dynamic cross-chain consensus mechanism to ensure secure data exchange; and (iii) developing a modular verification method to guarantee data integrity and support real-time verification.

TEACHING EXPERIENCE

Data Mining <i>Guest lecturer</i> Aalborg University	Spring 2025 Aalborg, Denmark
Algorithms and Data Structures <i>Guest lecturer</i> Aalborg University	Fall 2025 Aalborg, Denmark
Object-Oriented Programming <i>Teaching Assistant</i> Northeastern University	Spring 2020, 2021, 2022, 2023 Shenyang, China
Programming Practice Course <i>Teaching Assistant</i> Northeastern University	Spring 2020, 2021, 2022 Shenyang, China
Software Engineering <i>Teaching Assistant</i> Northeastern University	Spring 2020 Shenyang, China
Introduction to Big Data <i>Guest lecturer</i> Northeastern University	Spring 2021 Shenyang, China
Computer Networks <i>Guest lecturer</i> Northeastern University	Fall 2019 Shenyang, China

ACADEMIC SERVICE

Session Chair:

Organizer <i>IEEE Intelligent Conference on Intelligence and Security Informatics (ISI)</i>	Special Session
Topic: Trustworthy Multimodal, Generative, and Reinforcement Learning for Intelligent Systems	2026

Reviewer:

Program Committee <i>IEEE International Conference on Data Engineering (ICDE)</i>	2027
Program Committee <i>The AAAI Conference on Artificial Intelligence (AAAI)</i>	2026, 2027
Program Committee <i>International ACM Conference on Knowledge and Information Management (CIKM)</i>	2026
Program Committee <i>IEEE International Conference on Data Mining (ICDM)</i>	2026
Program Committee <i>APWeb-WAIM joint International Conference on Web and Big Data (APWeb-WAIM)</i>	2026
External Reviewer <i>IEEE International Conference on Data Mining (ICDM)</i>	2025
External Reviewer <i>APWeb-WAIM joint International Conference on Web and Big Data (APWeb-WAIM)</i>	2025
Reviewer <i>IEEE Transactions on Industrial Informatics (TII)</i>	2025, 2026
Reviewer <i>Distributed and Parallel Databases (DPD)</i>	2025
Reviewer <i>Computing</i>	2025
Reviewer <i>IEEE Network</i>	2025
Reviewer <i>The Journal of Supercomputing</i>	2025
Reviewer <i>World Wide Web: Internet and Web Information Systems (WWWJ)</i>	2024